

SOF IMO — Topic Deep-Dive: Division

For Ambar, Class III CBSE

1. What division means

Division is **sharing equally** or **making equal groups**.

- **Sharing:** 12 laddoos shared equally among 3 children → each child gets $12 \div 3 = 4$ laddoos.
- **Grouping:** 12 laddoos packed 3 per box → $12 \div 3 = 4$ boxes.

Both stories use the same fact, but the question changes: “how many EACH?” (sharing) vs “how many GROUPS?” (grouping). IMO papers use both wordings.

Division is also **repeated subtraction**: $12 - 3 - 3 - 3 - 3 = 0$, so 3 goes into 12 four times.

The names of the parts

In $14 \div 2 = 7$:

Number	Name	Meaning
14	Dividend	the number being divided
2	Divisor	the number we divide by
7	Quotient	the answer
(0)	Remainder	what is left over

2. Division and multiplication are a family

Every multiplication fact gives TWO division facts:

Multiplication	Division fact 1	Division fact 2
$6 \times 7 = 42$	$42 \div 6 = 7$	$42 \div 7 = 6$
$8 \times 9 = 72$	$72 \div 8 = 9$	$72 \div 9 = 8$

To solve $56 \div 8$, think: “8 times WHAT is 56?” → $8 \times 7 = 56$ → answer 7. Strong tables of 2–10 make division automatic. This is the single biggest score-booster.

3. Special rules

Rule	Example
Any number $\div 1$ = the number itself	$7 \div 1 = 7$
Any number $\div$ itself = 1	$9 \div 9 = 1$
$0 \div$ any number = 0	$0 \div 5 = 0$
A number can NEVER be divided by 0	$5 \div 0$ has no answer

4. Remainders

When things do not share out exactly, the leftover is the **remainder**.

Worked example: $17 \div 5$. Think tables of 5: $5 \times 3 = 15$ ($5 \times 4 = 20$ is too big). Quotient = 3. Leftover: $17 - 15 = 2$. So $17 \div 5 = 3$ remainder 2.

Rule to check yourself: the remainder must always be SMALLER than the divisor.

The “round up or round down?” trap

Word problems with remainders come in two flavours:

- “26 eggs, trays of 6. How many trays are **needed to hold all** the eggs?” → 4 trays hold 24, 2 eggs remain → need **5** trays (round UP).
- “65 books, shelves of 9. How many shelves are **completely full**?” → $9 \times 7 = 63$ → **7** full shelves (ignore the 2 leftover books).

Read the question, not just the numbers.

5. Common traps

Trap	Wrong thinking	Right thinking
Subtracting instead of dividing	$24 \div 6 = 18$	$24 \div 6 = 4$
Forgetting the remainder	$26 \div 6 = 4$ trays “enough”	4 trays + 2 left → 5 needed
Mixing up dividend and divisor	reading $54 \div 9$ as $9 \div 54$	big number first here
Remainder bigger than divisor	$19 \div 3 = 5 \text{ r } 4$	$19 \div 3 = 6 \text{ r } 1$
Quotient vs divisor names	calling the answer the divisor	answer = quotient

6. High-frequency IMO question types

1. Direct facts: “ $48 \div 6 = \underline{\quad}$ ” (tables 2–10).
2. Missing number: “ $\underline{\quad} \div 4 = 8$ ” (multiply back: $8 \times 4 = 32$).
3. Remainder: “What is the remainder when 22 is divided by 4?”
4. Sharing word problem: money, laddoos, marbles among friends.
5. Grouping word problem: teams, boxes, rows, trays.
6. Two-step Achiever’s: “3 packets of 8 biscuits shared among 4 friends” (multiply first, then divide).
7. Fact-family: “ $9 \times 7 = 63$, so which is true?”

7. Key facts checklist

- ☐ Tables 2–10 fluent both ways ($6 \times 7 = 42$ AND $42 \div 7 = 6$)
- ☐ Division = equal sharing = equal grouping = repeated subtraction
- ☐ Dividend $\div$ Divisor = Quotient (+ Remainder)
- ☐ $n \div 1 = n$, $n \div n = 1$, $0 \div n = 0$
- ☐ Remainder is always smaller than the divisor
- ☐ “Trays needed” rounds up; “completely full” rounds down

8. Quick drill (10 questions)

1. $36 \div 6 = ?$
2. $63 \div 7 = ?$
3. $8 \div 8 = ?$
4. What is the remainder when 20 is divided by 3?
5. $\underline{\quad} \div 5 = 6$. Missing number?
6. 40 bananas shared among 8 monkeys. Each gets?
7. 27 children make rows of 9. How many rows?
8. 31 pencils, boxes of 4. How many FULL boxes?
9. Riya has 2 packets of 10 biscuits. She shares all among 5 friends. Each gets?
10. $7 \times 6 = 42$. So $42 \div 6 = ?$

Answers: 1) 6 2) 9 3) 1 4) 2 ($3 \times 6 = 18$, $20 - 18 = 2$) 5) 30 6) 5 7) 3 8) 7 (28 pencils; 3 left over) 9) 4 ($20 \div 5$) 10) 7